

## OCEANSIDE, CA, USA

WMO: 722934

Lat: 33.219N

Lon: 117.349W

Elev: 9

StdP: 101.22

Time Zone: -8.00 (NAP)

Period: 99-19

WBAN: 53121

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 12            | 1.4        | 2.6 | -7.9                          | 1.9 | 13.3 | -5.3 | 2.4 | 12.6 | 7.5                   | 14.6 | 6.7 | 15.2 | 0.9                   | 30   | 0.369 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 7.7                    | 29.8            | 18.1 | 27.9 | 18.8 | 26.4 | 19.0 | 21.8                | 26.5 | 21.2 | 25.5 | 20.6 | 24.7 | 3.8                  | 250  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 20.2                            | 14.9 | 24.4 | 19.5 | 14.2 | 23.9 | 18.9 | 13.7 | 23.4 | 63.9          | 26.4 | 61.4 | 25.8 | 59.3 | 24.8 | 24.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 6.4           | 5.6  | 5.1 | DB                         | -1.0 | 35.0               | 1.4 | 2.8                                                | -2.0 | 37.0       | -2.8 | 38.6       | -3.6 | 40.2       | -4.6 | 42.2 | (4) |
| (5)               |      |     | WB                         | -2.4 | 23.1               | 1.6 | 0.9                                                | -3.5 | 23.7       | -4.5 | 24.2       | -5.4 | 24.7       | -6.6 | 25.3 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 16.2  | 12.1  | 12.3  | 13.6  | 14.9  | 16.9  | 18.7  | 20.9  | 21.1  | 20.2  | 17.6  | 14.4  | 11.7 | (6)  |
| (7)  |                                                                                  | DBStd       | 3.97  | 2.49  | 2.45  | 2.21  | 2.07  | 1.80  | 1.39  | 1.66  | 1.64  | 2.38  | 2.27  | 2.41  | 2.30 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 24    | 8     | 6     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 9    | (8)  |
| (9)  |                                                                                  | HDD18.3     | 1055  | 194   | 171   | 148   | 107   | 51    | 11    | 1     | 1     | 6     | 41    | 119   | 206  | (9)  |
| (10) |                                                                                  | CDD10.0     | 2292  | 72    | 69    | 112   | 146   | 214   | 260   | 337   | 343   | 306   | 237   | 134   | 61   | (10) |
| (11) |                                                                                  | CDD18.3     | 281   | 0     | 1     | 1     | 2     | 6     | 22    | 79    | 86    | 62    | 19    | 3     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 1489  | 24    | 27    | 39    | 68    | 75    | 42    | 229   | 306   | 346   | 233   | 92    | 7    | (12) |
| (13) | CDH26.7                                                                          | 321         | 3     | 4     | 9     | 21    | 29    | 7     | 22    | 37    | 89    | 74    | 24    | 1     | (13) |      |
| (14) | Wind                                                                             | WSAvg       | 1.9   | 1.6   | 1.9   | 1.9   | 2.1   | 2.2   | 2.1   | 2.0   | 1.9   | 1.9   | 1.7   | 1.6   | 1.6  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 287   | 58    | 59    | 40    | 22    | 7     | 3     | 3     | 1     | 4     | 14    | 22    | 42   | (15) |
| (16) |                                                                                  | PrecMax     | 584   | 252   | 255   | 160   | 53    | 29    | 20    | 43    | 9     | 28    | 126   | 76    | 203  | (16) |
| (17) |                                                                                  | PrecMin     | 90    | 4     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (17) |
| (18) |                                                                                  | PrecStd     | 136   | 62    | 56    | 39    | 16    | 8     | 5     | 9     | 2     | 6     | 27    | 21    | 42   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 26.2  | 27.0  | 27.5  | 30.1  | 30.4  | 27.3  | 28.9  | 29.7  | 32.8  | 31.9  | 29.5  | 23.9 | (19) |
| (20) |                                                                                  |             | MCWB  | 13.3  | 14.2  | 15.0  | 15.7  | 14.8  | 19.7  | 21.3  | 20.9  | 21.2  | 17.6  | 15.0  | 12.1 | (20) |
| (21) |                                                                                  | 2%          | DB    | 23.0  | 23.0  | 23.4  | 25.0  | 24.5  | 24.2  | 26.9  | 27.5  | 28.9  | 28.7  | 26.1  | 21.5 | (21) |
| (22) |                                                                                  |             | MCWB  | 12.4  | 12.8  | 14.1  | 14.8  | 17.0  | 19.0  | 21.1  | 21.1  | 20.7  | 16.3  | 14.2  | 12.2 | (22) |
| (23) |                                                                                  | 5%          | DB    | 20.9  | 20.2  | 20.5  | 21.8  | 22.1  | 23.0  | 25.7  | 26.2  | 26.9  | 25.7  | 22.8  | 19.3 | (23) |
| (24) |                                                                                  |             | MCWB  | 12.4  | 12.8  | 13.9  | 15.1  | 16.8  | 18.2  | 20.6  | 20.6  | 20.5  | 16.8  | 13.6  | 12.7 | (24) |
| (25) |                                                                                  | 10%         | DB    | 18.1  | 17.9  | 18.6  | 19.9  | 20.8  | 22.2  | 24.5  | 25.0  | 25.0  | 23.4  | 20.9  | 17.7 | (25) |
| (26) |                                                                                  |             | MCWB  | 12.2  | 12.9  | 14.0  | 14.8  | 16.0  | 17.7  | 19.9  | 20.0  | 19.6  | 17.5  | 14.4  | 12.6 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 16.1  | 16.4  | 16.9  | 18.3  | 19.0  | 20.3  | 22.6  | 22.5  | 22.9  | 21.3  | 17.8  | 16.7 | (27) |
| (28) |                                                                                  |             | MCDWB | 19.9  | 21.2  | 22.2  | 25.3  | 24.2  | 25.5  | 26.5  | 27.5  | 29.1  | 25.3  | 21.1  | 19.1 | (28) |
| (29) |                                                                                  | 2%          | WB    | 15.1  | 15.1  | 15.8  | 16.7  | 17.9  | 19.4  | 21.6  | 21.6  | 21.8  | 20.0  | 16.9  | 15.4 | (29) |
| (30) |                                                                                  |             | MCDWB | 18.4  | 18.5  | 19.8  | 21.4  | 22.1  | 23.8  | 25.5  | 26.1  | 27.1  | 24.4  | 20.4  | 17.7 | (30) |
| (31) |                                                                                  | 5%          | WB    | 14.3  | 14.4  | 15.1  | 15.7  | 17.0  | 18.7  | 20.9  | 20.9  | 21.0  | 18.9  | 16.2  | 14.5 | (31) |
| (32) |                                                                                  |             | MCDWB | 17.6  | 18.0  | 18.8  | 20.0  | 20.8  | 22.6  | 24.9  | 25.3  | 25.4  | 22.9  | 20.1  | 17.1 | (32) |
| (33) |                                                                                  | 10%         | WB    | 13.5  | 13.6  | 14.5  | 15.1  | 16.4  | 18.0  | 20.4  | 20.4  | 20.2  | 18.0  | 15.4  | 13.7 | (33) |
| (34) |                                                                                  |             | MCDWB | 16.9  | 17.2  | 17.8  | 19.0  | 19.8  | 21.3  | 24.1  | 24.4  | 24.1  | 21.6  | 19.5  | 16.5 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 13.2  | 11.9  | 10.4  | 10.2  | 8.1   | 7.1   | 7.2   | 7.7   | 9.3   | 11.5  | 13.4  | 13.1 | (35) |
| (36) |                                                                                  |             | MCDBR | 17.7  | 17.2  | 15.3  | 16.1  | 11.7  | 8.9   | 8.6   | 9.6   | 12.2  | 17.9  | 18.3  | 16.3 | (36) |
| (37) |                                                                                  |             | MCWBR | 9.9   | 9.7   | 8.7   | 8.7   | 6.2   | 4.6   | 4.1   | 4.7   | 6.0   | 9.7   | 10.2  | 10.0 | (37) |
| (38) |                                                                                  | 5% WB       | MCDWB | 12.6  | 12.6  | 10.9  | 11.7  | 9.5   | 8.7   | 8.2   | 8.7   | 10.5  | 12.5  | 13.4  | 11.4 | (38) |
| (39) |                                                                                  |             | MCWBR | 8.1   | 8.0   | 6.9   | 6.9   | 5.2   | 4.6   | 4.0   | 4.4   | 5.4   | 7.6   | 8.6   | 8.0  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.310 | 0.320 | 0.338 | 0.352 | 0.371 | 0.368 | 0.396 | 0.391 | 0.362 | 0.357 | 0.317 | 0.319 | (40) |      |
| (41) |                                                                                  | taud        | 2.568 | 2.553 | 2.485 | 2.452 | 2.426 | 2.440 | 2.434 | 2.430 | 2.498 | 2.482 | 2.580 | 2.573 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 914   | 937   | 941   | 935   | 915   | 912   | 885   | 887   | 903   | 881   | 897   | 881   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 80    | 90    | 104   | 112   | 116   | 114   | 115   | 113   | 101   | 95    | 79    | 76    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 3.26  | 4.13  | 5.46  | 6.59  | 6.98  | 7.36  | 7.29  | 6.99  | 5.96  | 4.57  | 3.57  | 2.95  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.21  | 0.38  | 0.29  | 0.35  | 0.49  | 0.46  | 0.40  | 0.21  | 0.26  | 0.32  | 0.22  | 0.21  | (45) |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |      |  |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|------|--|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |      |  |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |      |  |
| (46) | Station Only            | +0.56 | N/A     | N/A    | +1.54   | N/A   | N/A   | N/A         | -121    | +214    | +110    | (46) |  |
| (47) | Regional (47 neighbors) | +0.61 | N/A     | -1.67  | +0.66   | N/A   | N/A   | N/A         | -103    | +217    | +112    | (47) |  |

Nomenclature: See separate page